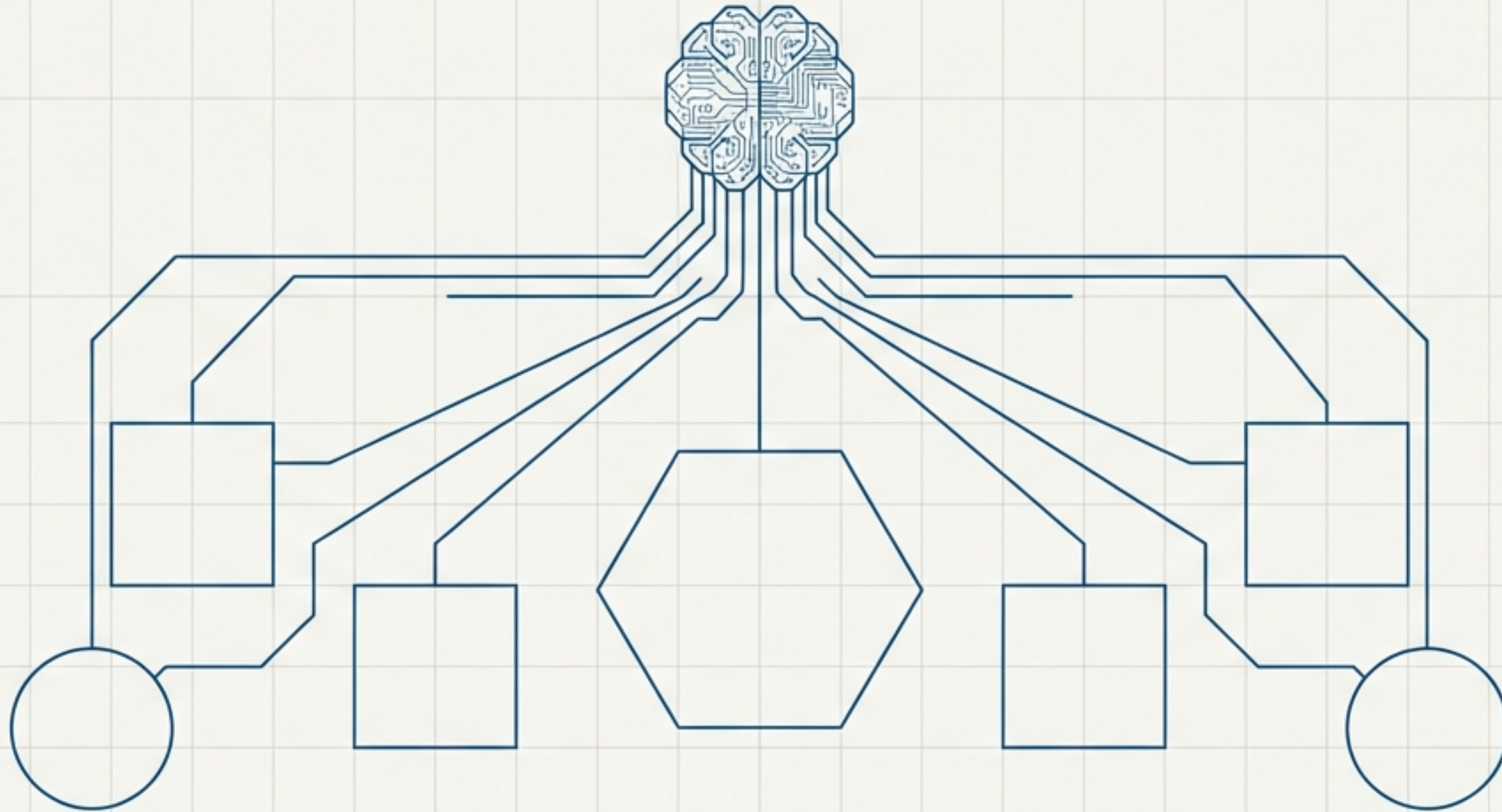


Phase 2: An Architectural Blueprint for Domain Expansion

Building the Body



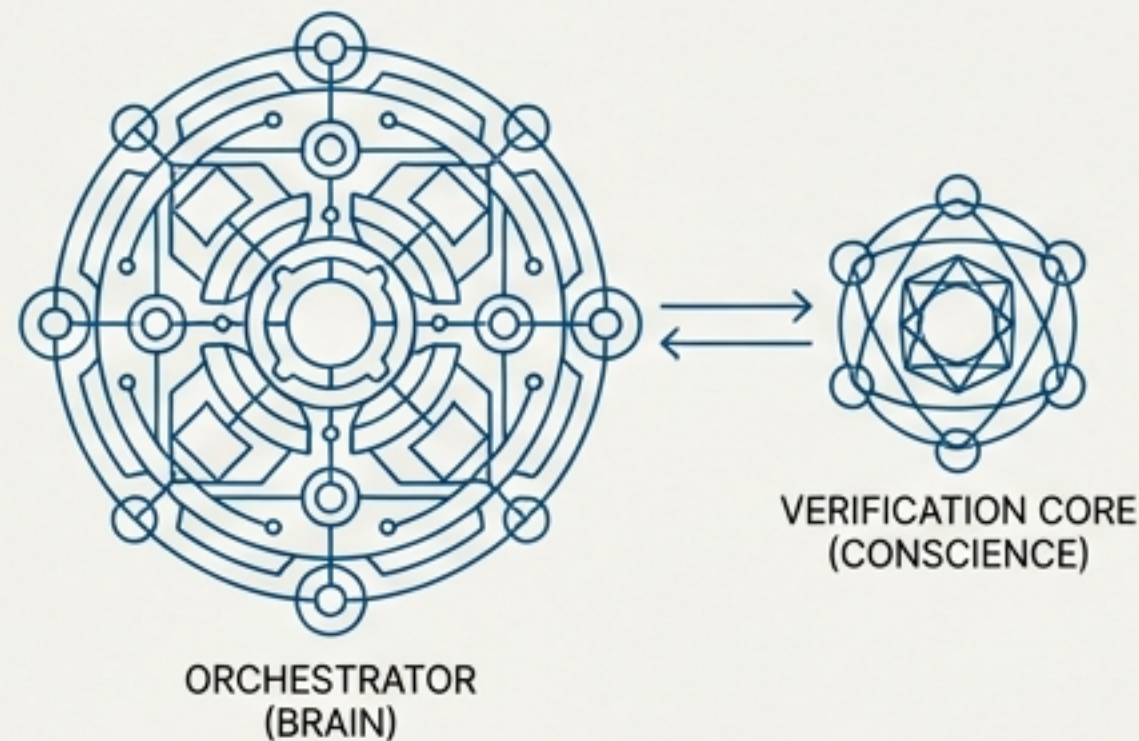
The Mandate: From Central Nervous System to Specialized Body

Phase 1 Accomplished: The Brain & Conscience

Established the Tier 1 **Orchestrator** (The Central Nervous System).

Built the **Verification Core** (The Conscience).

Result: A system that can manage and verify tasks.

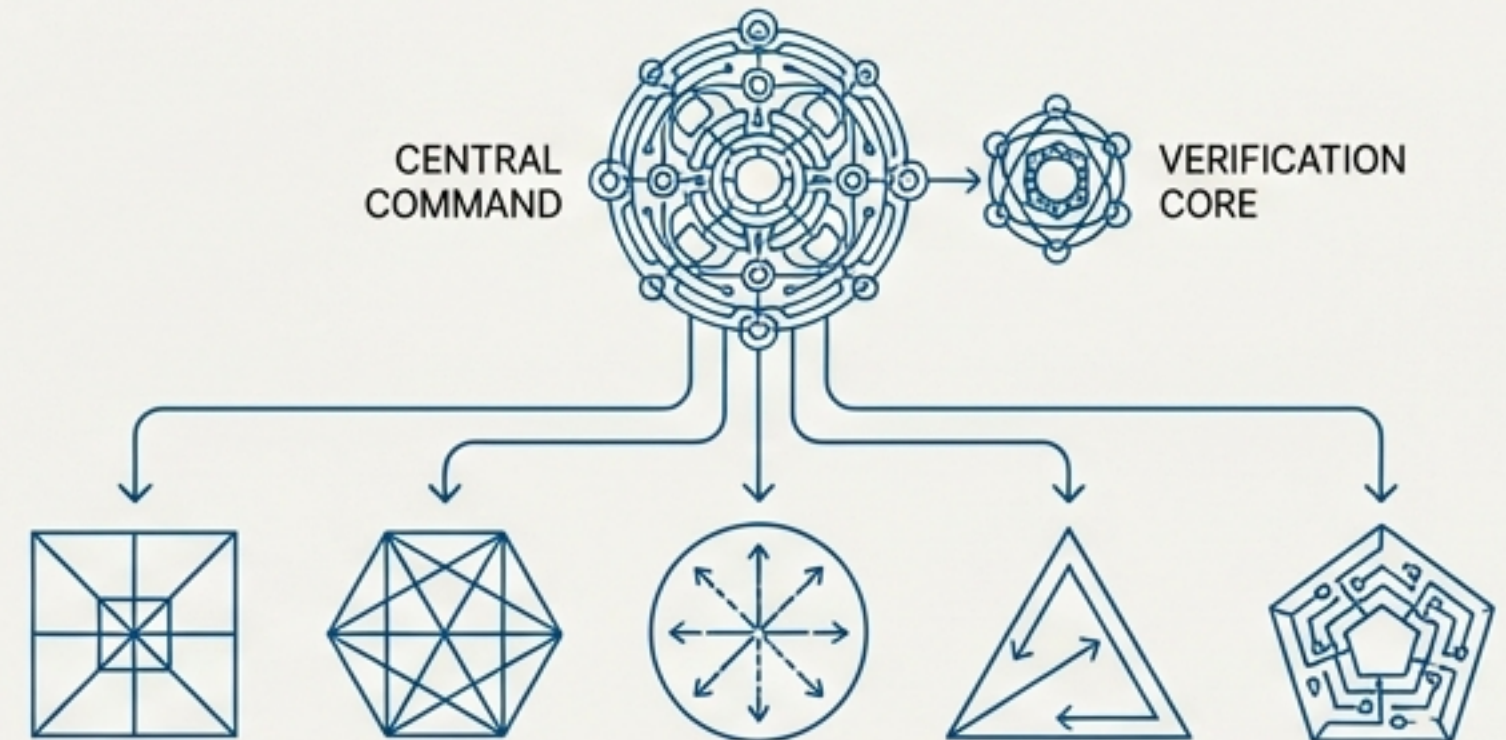


Phase 2 Objective: The Body

Our mission is **Massive Domain Expansion**.

We will construct a powerful, specialized workforce of mathematical agents.

Result: A system that can execute complex, domain-specific tasks.



The Core Challenge: To prevent the 'Jack of all trades, master of none' failure mode.

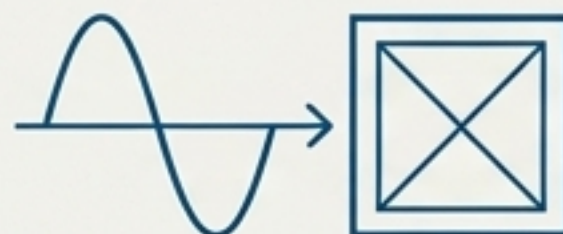
The Strategy: Achieve this through radical **Domain Decomposition**.

The Four Non-Negotiable Architectural Principles



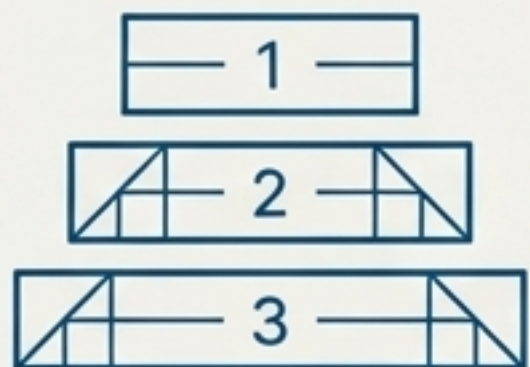
Embrace Granular Specialization

Avoid a single, monolithic 'Math Agent.' We will create a collective of highly specialized sub-teams, each tasked with a narrow, specific mathematical function to ensure maximum accuracy and efficiency.



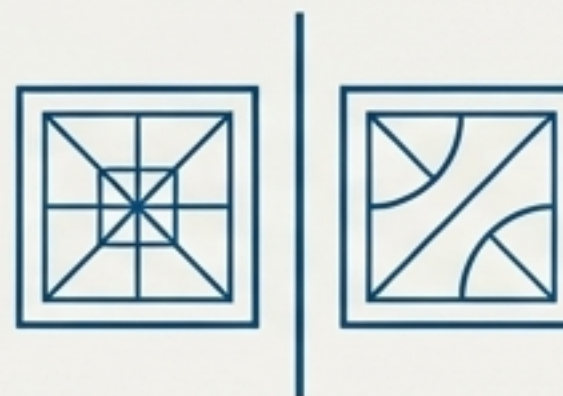
Prioritize Algorithmic Wrappers

Guarantee mathematical precision by wrapping specific, deterministic algorithms (e.g., Risch, GMP). LLM-based reasoning is for orchestration and strategy, not for raw computation.



Adhere to the Tiered Hierarchy

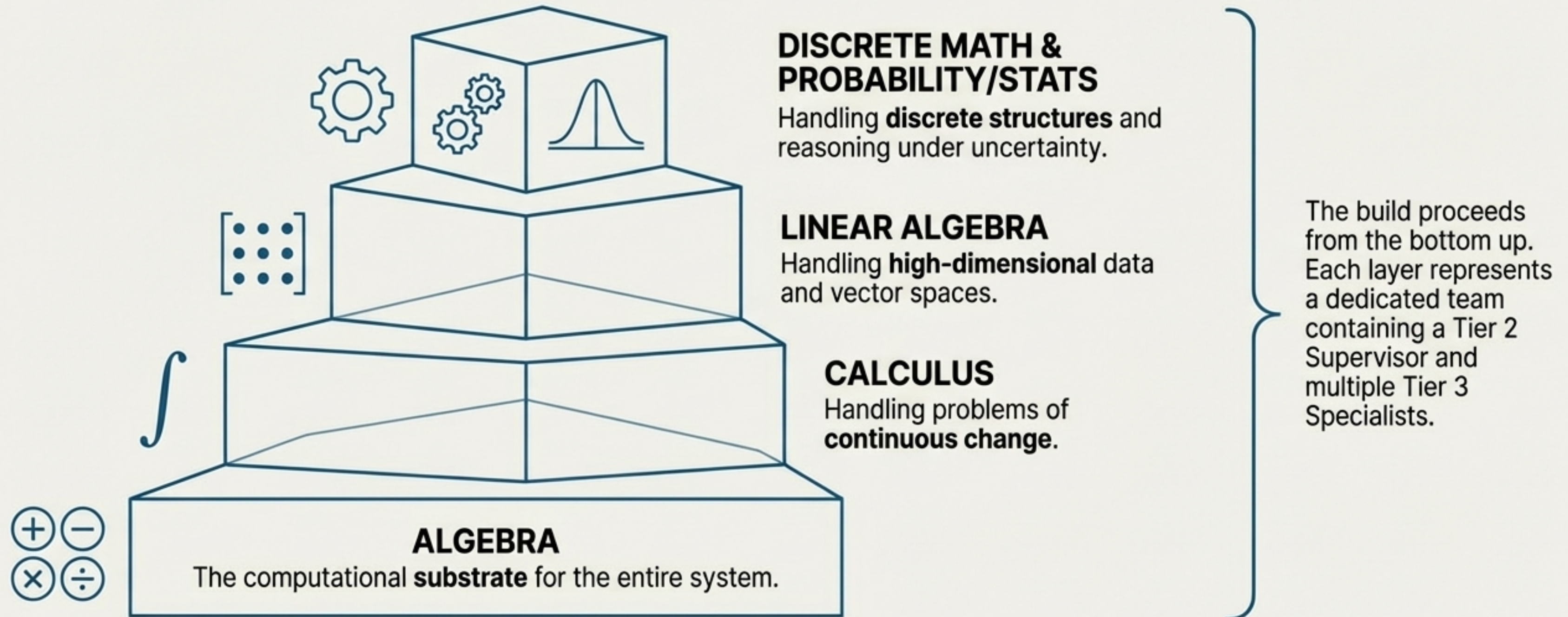
Maintain a clear structure: Tier 1 (Orchestrator), Tier 2 (Supervisors for major domains), and Tier 3 (Specialists executing algorithms) to manage complexity and delegate responsibility.



Enforce Strict Separation of Concerns

Prevent logical and algorithmic conflicts. Mandatory separations include **Symbolic (exact) vs. Numerical (approximate)** methods in Calculus, and **Bayesian vs. Frequentist** reasoning in Statistics.

The Build Plan: A Layered Agent Hierarchy



Layer 1: The Algebra Team (The Foundation)



To establish the computational substrate. If this layer fails to simplify a polynomial correctly, the Calculus team cannot integrate it.

Agent Designation	Core Directive	Technical Specification & Key Algorithms
Algebra Supervisor (Tier 2)	Act as the foreman. Focus on simplification strategy and delegation.	Understands task prerequisites (e.g., factoring before finding roots) and routes to the appropriate specialist.
Arithmetic Specialist	Perform Arbitrary-Precision Arithmetic .	Must wrap libraries like GMP or mpmath . Standard floating-point math is strictly forbidden.
Polynomial Manipulation Specialist	Specialize in the structure of polynomials.	Implements wrappers for Gröbner Bases (e.g., Buchberger's algorithm) to solve systems of polynomial equations.
Number Theory Specialist	Manage tasks related to discrete integers.	Capabilities: Primality testing (Miller-Rabin), Integer Factorization (Pollard's rho), Diophantine equations.

Layer 2: The Calculus Team (The Analysis Layer) $\int$

To handle continuous change, requiring the strictest separation between Symbolic (exact) and Numerical (approximate) methods.

Agent Designation	Core Directive	Technical Specification & Key Algorithms
Calculus Supervisor (Tier 2)	Act as gatekeeper with crucial decision logic to route tasks correctly.	Must distinguish between requests for exact solutions (Symbolic) and approximate solutions (Numerical).
Differentiation Specialist	Compute derivatives for single and multi-variable functions.	Capable of calculating gradients, Jacobians, and Hessians.
Integration Specialist (Critical Split)	Provide exact and approximate integrals via a dual-engine approach.	Symbolic Engine: Wraps a Risch Algorithm solver.
		Numerical Engine: Falls back to Quadrature methods.
Differential Equation Solver	Solve Ordinary and Partial Differential Equations (ODEs/PDEs).	Selects between exact symbolic matches and numerical solvers like Runge-Kutta .
Series Specialist	Specialize in series expansions and convergence analysis.	Dedicated to Taylor and Fourier series expansions to approximate functions.

Layer 3: The Linear Algebra Team (The Vector Layer)

Objective: To handle high-dimensional data, where computational stability is paramount and rounding errors can compound catastrophically.

Agent Designation	Core Directive	Technical Specification & Key Algorithms
Linear Algebra Supervisor (Tier 2)	Analyze tasks and select optimal algorithms based on matrix properties.	Must identify properties like sparse vs. dense, symmetric, and positive-definite to ensure stability.
Matrix Operations Specialist	Handle fundamental matrix arithmetic and property calculations.	Responsible for matrix addition/multiplication, traces, determinants, and transposes.
Decomposition Specialist	Perform the "heavy lifting" of matrix factorizations.	Must support SVD (Singular Value Decomposition) , QR , LU , and Cholesky decompositions.
Vector Space Analyst	Calculate abstract properties of matrices and vector spaces.	Bridges computation and theory by calculating Basis, Rank, Nullspace, and Eigenvalues/Eigenvectors.

Layer 4: The Logic & Uncertainty Teams



The Discrete Math Team

Objective: Handle discrete structures using rigid algorithms and structured data representations.

- **Combinatorics Agent:** Handles counting and permutations. Requires algorithmic backing (e.g., generating functions) as this is a known weakness for generative AI.
- **Graph Theory Agent:** Wraps canonical graph algorithms (Dijkstra, BFS/DFS). Requires structured input like an adjacency matrix, not raw text.

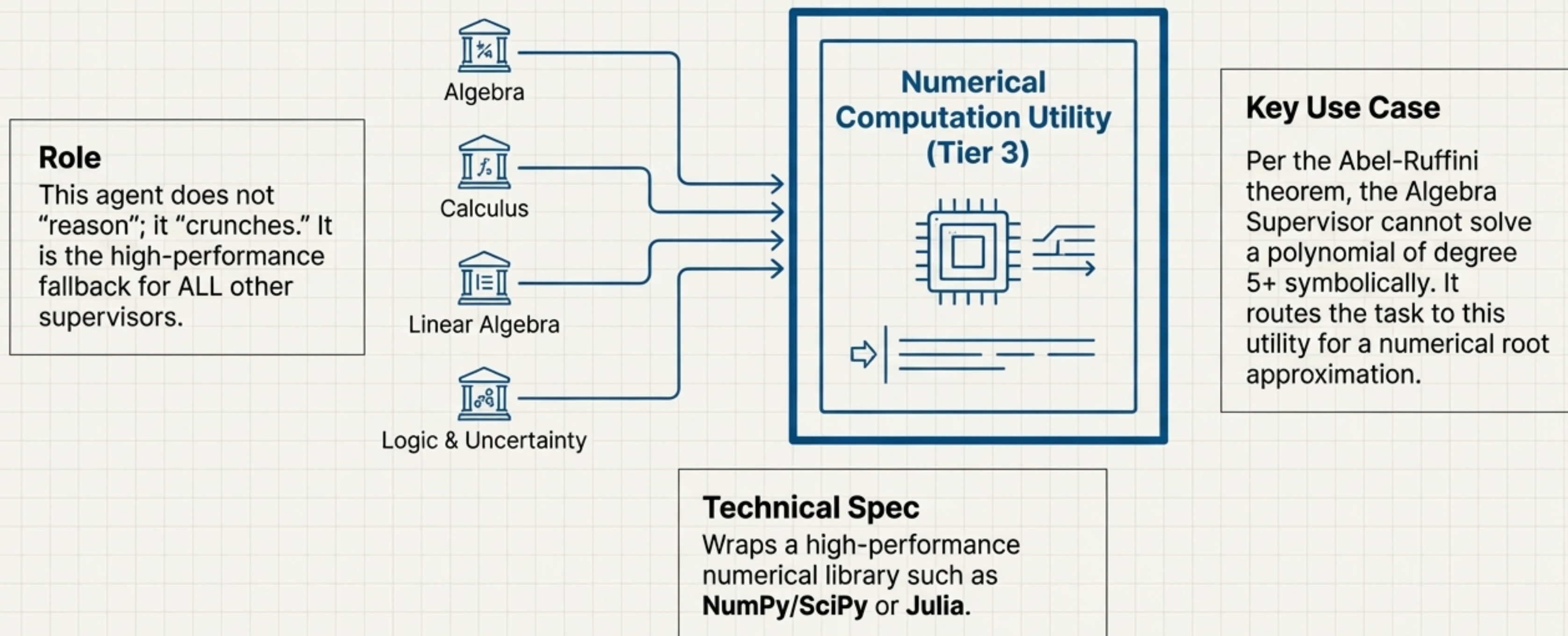
The Probability & Statistics Team

Objective: Handle reasoning under uncertainty with a strict philosophical separation of methodologies.

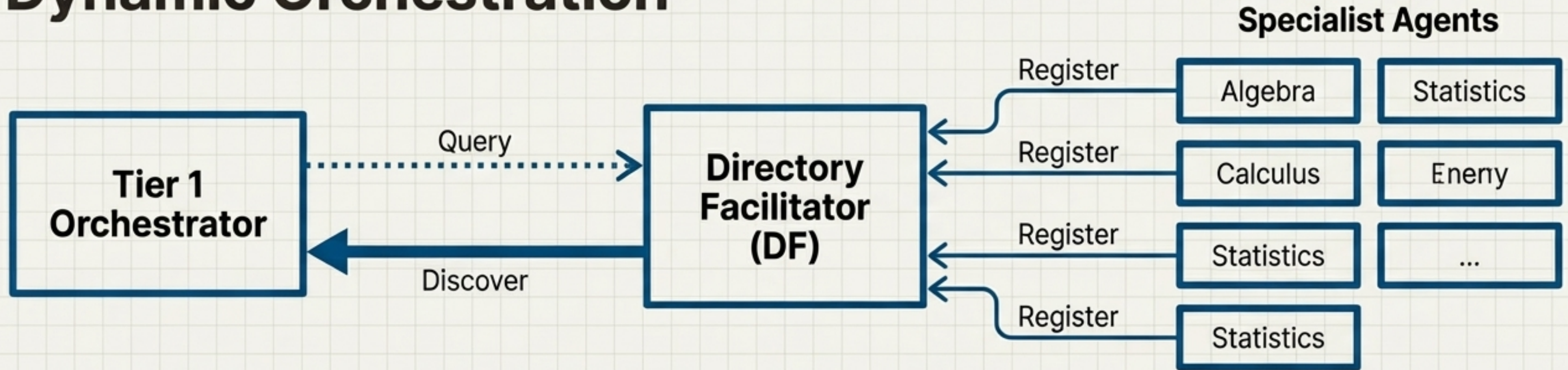
- **Prob & Stats Supervisor:** Enforces the split. Routes 'confidence interval' requests to the Frequentist agent and 'posterior update' requests to the Bayesian agent.
- **Bayesian Inference Engine:** Dedicated agent for posterior calculations using **Markov Chain Monte Carlo (MCMC)**.
- **Frequentist Agent:** Handles classical hypothesis testing (t-tests, p-values) and confidence intervals.

The System Safety Net: The Numerical Computation Utility

Objective: To ensure the system can always provide a high-quality approximate answer when a symbolic or exact solution is computationally intractable or mathematically impossible.



Wiring the Body to the Brain: Dynamic Orchestration



Step 1: Directory Facilitator (DF) Registration

Every new Supervisor and Specialist **must** register its capabilities with the DF. This registration is precise, including service name and properties.

```
service: math.calculus.integration
properties: { type: symbolic, algorithm: risch }
```

Step 2: Dynamic Orchestration

Because the Tier 1 Orchestrator queries the DF in real-time, it automatically 'sees' and can delegate tasks to the newly created agents. The Orchestrator's capabilities expand organically as new tools are added to its 'rolodex,' fulfilling the architectural promise of a modular, scalable system.

Phase 2 Definition of Done: The “Fragile Genius”



Neue Haas Grotesk Display Pro (Bold) The Genius

The system now possesses immense and verifiable mathematical power.

Key Capabilities Achieved:

- Executing complex Singular Value Decompositions (SVDs).
- Finding closed-form integrals for complex functions.
- Solving problems in number theory using deterministic algorithms.

Neue Haas Grotesk Display Pro (Bold) The Fragility

The system is powerful but brittle. Its specialized agents lack robust precondition validation and meta-cognitive oversight.

Identified Risks:

- **No Precondition Validation:** A Supervisor routing a negative number to a Logarithm Specialist will cause a component crash.
- **No Conflict Resolution:** If symbolic and numerical agents produce conflicting results, there is no higher-level mechanism to adjudicate the disagreement.

From Fragility to Robustness: The Mandate for Phase 3

The fragility of the Phase 2 deliverable is an expected and planned outcome. It explicitly necessitates the next major development cycle:

Phase 3: Implementing the Meta-Cognitive Middleware.

Key Objectives for Phase 3:

- **Precondition Validation:** Build a systemic error-handling layer to intercept invalid inputs before they reach specialists.
- **Knowledge Management & Conflict Resolution:** Create systems to adjudicate disagreements between agents and manage shared knowledge.

In Phase 3, we will stabilize this powerful workforce, transforming the 'Fragile Genius' into a robust, self-correcting intellect.